



# Cloudflare Radar 2024 Year in Review

Global Internet traffic, adoption shifts, disruptions, and security trends  
observed through Cloudflare's network in 2024

For technology executives and network infrastructure leaders

# 17% Global Traffic Growth and Starlink's 3.3x Surge

Global Internet Traffic (YoY)

**+17%**

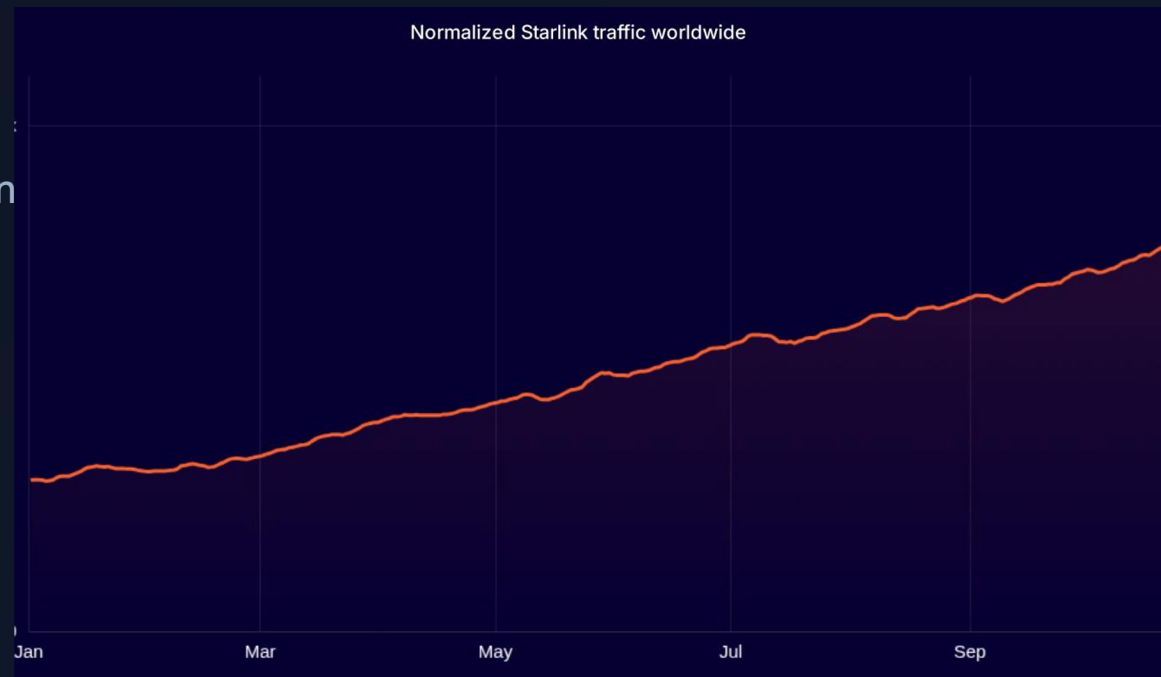
SpaceX Starlink Traffic Growth

**3.3x**

## Executive Insights

- Sustained global demand for Internet capacity across region
- Satellite broadband scaling rapidly in underserved areas
- Infrastructure planning should include non-terrestrial paths

Normalized Starlink traffic worldwide



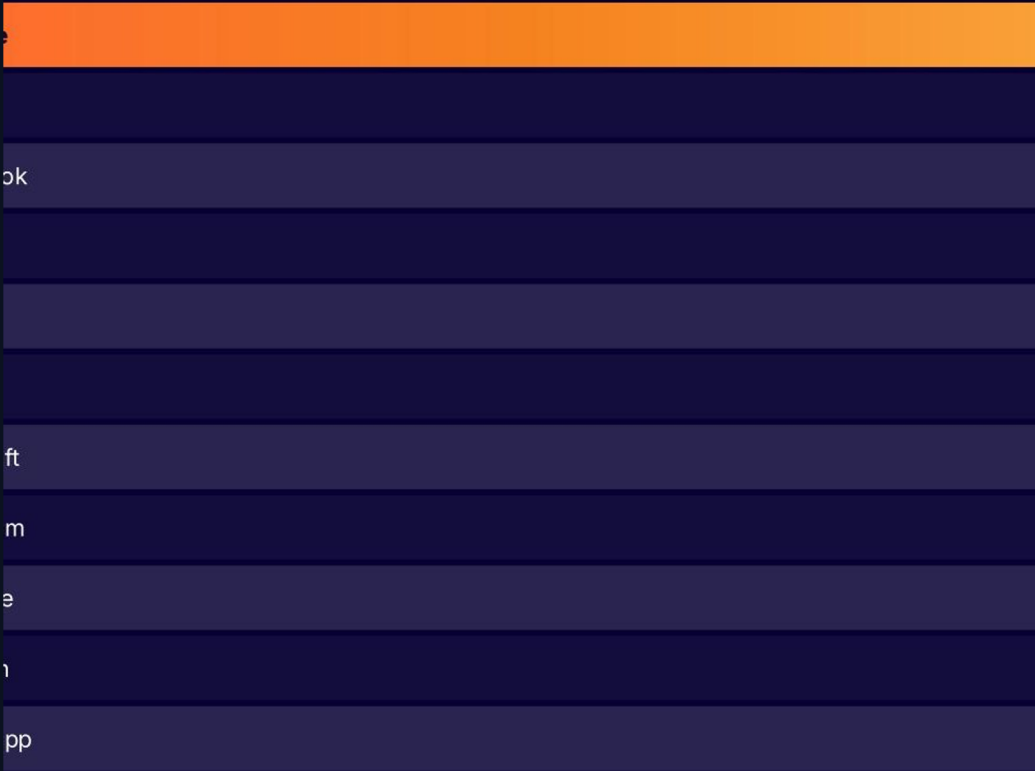
# Google Leads Top 10 Services as Generative AI Rises

## Top 10 Internet Services (2024)

- 1. Google 6. Microsoft
- 2. Facebook 7. Instagram
- 3. Apple 8. YouTube
- 4. TikTok 9. X/Twitter
- 5. Amazon 10. WhatsApp

## Generative AI Trend

Cloudflare Radar observed a continued meteoric rise in generative AI traffic and usage throughout 2024.



# Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3



## Adoption Metrics

Post-Quantum TLS 1.3 Share

**~2% → ~13%**

HTTP/3 Adoption

**21%**

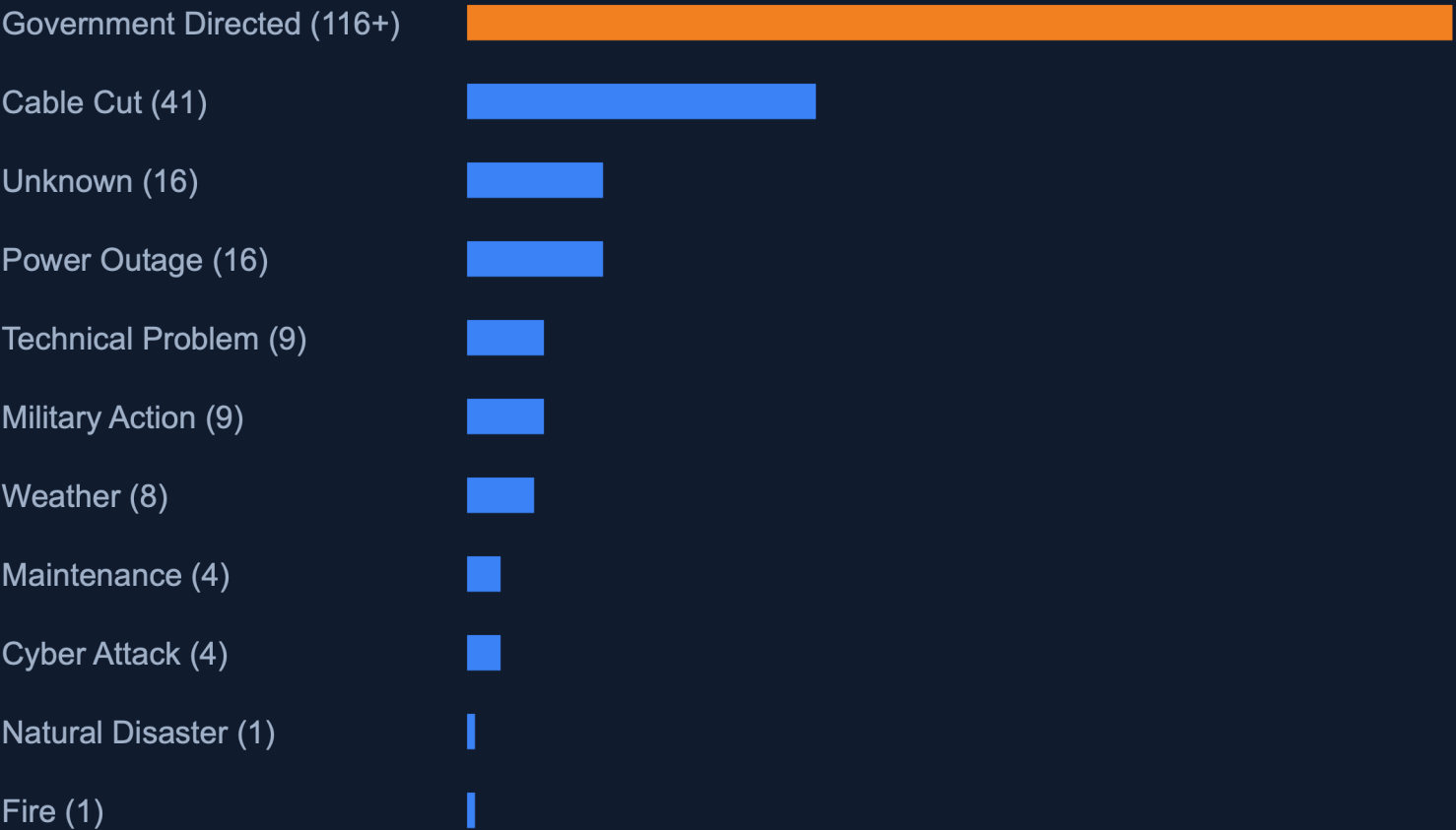
Top API Client Language

**Go (~12%)**

Inflection point: Chrome 124 enabled PQ by default (Apr 2)

# 225 Major Disruptions: Government Shutdowns Lead Outage Causes

## Cause Breakdown (Count)



Total: 225

Government-directed shutdowns were the leading disruption cause in 2024.



# 6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Traffic Mitigated

**6.5%**

Most Attacked Sector

**Gambling and Games**

Bot Concentration

**68%**

of global bot traffic from top 10 countries

## Persistent Threat Signals

- Log4j (CVE-2021-44228) still actively targeted
- Malicious emails: 4.3%
- Deceptive links in malicious emails: 43%
- .bar generated largest malicious/spam share

# Five Takeaways Shaping the 2025 Internet Outlook

1. Internet demand remains strong: +17% global growth; Starlink at 3.3x.
2. Post-quantum transition accelerated: TLS 1.3 PQ share ~2% to ~13%.
3. Connectivity resilience is geopolitical: shutdowns remain primary disruption driver.
4. Security pressure persists: 6.5% mitigated traffic; Log4j still exploited.
5. Stack modernization continues: HTTP/3 at 21%, Go leads API clients (~12%).

**2025 priority: resilience, cryptographic agility, and proactive threat operations.**